

Multi-Organ Ultrasound Image Analysis: A Dual-Model Approach for UUSIC Challenge

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Abstract

This technical report presents our approach for the Universal UltraSound Image Challenge (UUSIC), focusing on multi-organ classification and segmentation tasks. We implemented a dual-model framework consisting of separate networks for segmentation and classification tasks across seven different organs. Our method utilises standard UNet architecture for segmentation and EfficientNet for classification to address the multi-organ ultrasound

1 Introduction

The Universal UltraSound Image Challenge presents a unique multi-organ, multi-task scenario requiring algorithms to perform both classification and segmentation across diverse anatomical regions, including breast, thyroid, liver, kidney, fetal head, heart, and appendix. Each organ is associated with distinct clinical tasks, exemplifying the complexity of developing generalised machine learning models for ultrasound image analysis. Our approach addresses this challenge through a straightforward dual-model framework that separates segmentation and classification tasks.

2 Methodology

2.1 Overall Framework

Given the heterogeneous nature of the UUSIC challenge, we implemented a dual-model approach that separates segmentation and classification tasks. This design decision was motivated by the fundamental differences in the computational requirements and

feature representations needed for pixel-level segmentation versus image-level classification. Our framework consists of two specialised networks: a UNet-based architecture handling all segmentation tasks across different organs, and an EfficientNet-based architecture managing all classification tasks. This separation allows for task-specific optimisation while maintaining the ability to process multiple organ types within each task category. The segmentation model addresses four distinct regions of anatomical interest, including thyroid nodule, kidney, foetal head, and cardiac chambers. Meanwhile, the classification model handles three primary classification tasks encompassing breast nodule malignancy classification, fatty liver classification, and appendicitis classification, with additional support for breast cancer subtype classification.

2.2 Network Architectures

Our segmentation network employs a standard UNet architecture configured for medical ultrasound imaging [1]. The network features a symmetric encoder-decoder structure with skip connections, processing ultrasound images and producing binary segmentation masks. The encoder pathway utilises four levels with feature channel dimensions of 16, 32, 64, and 128, respectively, employing convolution layers with a stride of 2 for downsampling across three stages. Each level incorporates three residual blocks to enhance feature learning and gradient flow. The decoder pathway mirrors the encoder structure, progressively upsampling feature maps while integrating skip connections to preserve spatial details crucial for accurate boundary delineation. For classification tasks, we implemented an EfficientNet-B0 backbone architecture pretrained on ImageNet [2], adapted for multi-organ ultrasound classification. The network processes the ultrasound images and outputs a 10-dimensional vector representing different classification categories across multiple organs.

2.3 Data Processing Pipeline

All ultrasound images undergo standardised preprocessing to ensure consistent input characteristics across different organs and imaging conditions. Images are first normalised to the range from 0 to 1 through division by 255. Subsequently, all images are resized to a standard dimension of 224×224 pixels using bilinear interpolation for images and nearest-neighbour interpolation for segmentation masks to preserve label integrity. During training, we implement stochastic data augmentation (Random horizontal and vertical flipping) to improve model generalisation across diverse imaging conditions and anatomical variations.

3 Training Strategy

3.1 Loss Functions and Optimisation

Both models utilise cross-entropy loss functions appropriate for their respective tasks. The segmentation model employs pixel-wise cross-entropy loss comparing predicted probability maps with ground truth binary masks. The classification model uses

standard cross-entropy loss for multi-class classification, computing the negative log-likelihood of the correct class predictions. Optimisation is performed using the AdamW optimiser with a batch size of 32, learning rate of 5×10^{-4} for 200 epochs.

4 Conclusion

Our dual-model approach provides an effective and practical solution for the multi-organ, multi-task nature of the UUSIC challenge. By leveraging established architectures with appropriate adaptations for ultrasound imaging, we achieve a balance between performance, interpretability, and computational efficiency. The separation of segmentation and classification tasks enables specialised optimisation while maintaining the capability to handle the diverse range of ultrasound imaging scenarios present in the challenge dataset. The modular design facilitates easy adaptation to new organs or tasks, establishing a robust foundation for clinical ultrasound image analysis applications. This approach demonstrates that practical engineering solutions utilising proven architectures can effectively address complex multimodal medical imaging challenges while maintaining clinical relevance and deployment feasibility.

References

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- [2] Tan, M., Le, Q.: Efficientnet: Rethinking model scaling for convolutional neural networks. In: International Conference on Machine Learning, pp. 6105–6114 (2019). PMLR